

## ANU Physics Academic Integrity Declaration Form

Name: chhaya ramnani Student ID: u8336188

Date: 6 sep 2026 Assessment Item: assignment 5

### **Collaboration with others**

You are encouraged to discuss problems with other students, however you must submit your own work. The minimum penalty for plagiarism is a fail on the assessment task, and all incidents of plagiarism are recorded on a student's academic record.

**If you have collaborated with others, please note these people here:**

---

### **Use of AI Tools**

You may not present material produced by generative AI as your own work, as this is an academic integrity breach. If you have used AI in this assessment, the use must be declared here:

☐ No AI tools were used for this assignment

☒ AI tools were used as described below:

AI Tools Used: gemini

**How AI was used (check all that apply):**

☐ Mathematical calculations ☒ Problem solving ☐ Code generation  
☐ Writing assistance ☐ Research ☐ Other: \_\_\_\_\_

Chat Evidence Required - you must include your complete AI conversation(s):

☒ Full chat transcript pasted below  
☐ Full chat screenshots/files attached  
☐ Shared link: <https://share.gemini.google/GtMfKdCjNxi8>

I declare that:

- ☒ This AI use declaration is complete and accurate
- ☒ I have included unedited records of all AI interactions
- ☒ I understand the physics concepts and can explain my work
- ☒ I remain responsible for all content in my submission

Signature:  Date: 6 sep 2026

# Assignment 5

September 6, 2026

Given:

- $m_{H^0} \approx 125 \text{ GeV}/c^2$
- $m_{Z^0} \approx 91 \text{ GeV}/c^2$
- $m_e = 5.11 \times 10^{-4} \text{ GeV}/c^2$
- $m_\gamma = 0$

## 1 Part 1

The maximum energy of the electron at LEP was

$$E = 104 \text{ GeV}.$$

The total relativistic energy of a particle is

$$E = \gamma mc^2.$$

rearranging:

$$\gamma = \frac{E}{mc^2}.$$

rest mass of the electron is

$$m_e = 5.11 \times 10^{-4} \text{ GeV}/c^2.$$

so, solving for  $\gamma$

$$\gamma = \frac{104 \text{ GeV}}{5.11 \times 10^{-4} \text{ GeV}} \\ \gamma \approx 2.04 \times 10^5.$$

therefore to 1% accuracy,

$$\boxed{\gamma \approx 2.04 \times 10^5}$$

(super close to the speed of light)